

Strattum — Open Lakehouse Benchmark

🇵🇹 Versão em português: [BENCHMARK-LAKEHOUSE.md](#)

From 2 operational databases to a knowledge graph, in minutes, on a laptop. 3 million records (PostgreSQL + MongoDB) → open lakehouse (DuckLake + Parquet/S3) → modeled *clean* layer → entity graph — measured end to end.

This document reports a **reproducible** benchmark of the Strattum platform running **entirely on a single MacBook (8 GB of RAM)**. The goal is to answer, with numbers, a simple question: *how fast and how cheaply can any company turn its own operational data into something useful?*

TL;DR

What	Volume	Time	Throughput	Peak RAM
Ingestion PostgreSQL → lake	2,000,000 rows	36.9 s	~54,200 rows/s	272 MB
Ingestion MongoDB → lake	1,000,500 docs	25.0 s	~40,500 docs/s	272 MB
Transformation (dbt/DuckLake)	3,000,500 rows	3.8 s	~795,000 rows/s	382 MB
Incremental sync (+150k)	150,000 rows	3.5 s	delta-only, 0 duplicates	267 MB
Knowledge graph	290,756 nodes · 300,000 edges	126 s ¹	~4,680 elem/s	250 MB (worker) · 302 MB (FalkorDB)

¹ Graph built on a **representative slice** (100k contracts + linked entities); build optimized from 362 s → 126 s during the benchmark itself (see §4). Ingestion and *clean* ran on the **full** volume.

3 million rows ingested + modeled in ~66 seconds, peaking at ~380 MB of RAM per process — on desk hardware, over 100% open formats (Parquet + DuckLake catalog), with no proprietary data warehouse and no cluster.

The scenario

A typical mid-size company has its data scattered: a **PostgreSQL** with the people registry, a **MongoDB** with contracts/cases, spreadsheets, an ERP. Nobody "does anything with data" because pulling it all together normally takes a months-long project, an expensive data warehouse and a dedicated team.

The benchmark simulates exactly that case with realistic synthetic data:

- **PostgreSQL** — **users** : 2,000,000 users (id, name, e-mail, national id, city, timestamps).

- **MongoDB — 2 collections:**
- `contracts` : 1,000,000 contracts, each with **two parties** (`processante` and `processado` , both referencing a user) + product, status and value;
- `products` : 500 products (id → name/category).
- **Increment:** +100,000 users and +50,000 contracts, to measure the cost of a next-day sync.

The target: a *clean* layer with **one users table** and **one contracts table** (already `JOIN` ing the two Mongo collections to bring in the product name), and a **graph** where each contract connects its two parties and its product.

Setup (deliberately modest)

Item	Value
Machine	MacBook, Apple Silicon, 8 vCPU, 8 GB RAM
Docker (Linux VM)	3.8 GB for all containers
Sources	PostgreSQL 16, MongoDB 7 (containers)
Lakehouse	DuckLake (Postgres catalog) + Parquet on MinIO (S3)
Transformation	dbt running native DuckDB/DuckLake
Graph	FalkorDB (entity graph)
Orchestration	Prefect

Nothing here is "benchmark hardware". It is what any dev has on their desk. That is precisely the point: **the open architecture scales down** — it runs on a laptop today and on a cluster tomorrow, without switching formats.

Results by stage

1) Ingestion (bootstrap — full load)

Each connector **streams** from the source (server-side cursor in Postgres, batched cursor in Mongo) and writes Parquet to the lake in micro-batches. That is why the **memory peak is constant (~272 MB) regardless of volume** — the full dataset is never loaded into RAM.

Source	Rows	Time	Throughput	Peak RAM	Parquet written
PostgreSQL <code>users</code>	2,000,000	36.9 s	54,200 r/s	272 MB	83 MB
MongoDB <code>contracts</code>	1,000,000	24.7 s	40,500 r/s	272 MB	85 MB
MongoDB <code>products</code>	500	0.2 s	—	—	36 KB
Total	3,000,500	~62 s	~48,500 r/s	272 MB	~168 MB

3 million rows become ~168 MB of compressed columnar Parquet — the base is immediately ready for direct analytical querying, with no copy into a warehouse.

2) Transformation — *clean* layer (dbt over DuckLake)

dbt reads `raw` and materializes three *clean* tables in the lake itself. `clean.contracts` **JOIN**s the two Mongo **collections** (contracts × products), normalizes types and brings the product name onto each contract.

Model	Rows	Time
<code>clean.products</code>	500	0.40 s
<code>clean.users</code>	2,000,000	3.23 s
<code>clean.contracts</code> (join)	1,000,000	3.44 s
Total	3,000,500	3.8 s

Result: **0 contracts without a product** (join intact), ~795k rows/s. DuckDB over Parquet transforms millions of rows in seconds — without moving data out of the lake.

3) Incremental sync (the "next day")

After +100k users and +50k contracts land in the sources, the same pipeline runs again. The incremental cursor reads **only the delta** (`updated_at > watermark`) and *merges* by primary key — nothing is reprocessed.

Source	Delta	Time	vs. bootstrap
PostgreSQL	100,000	2.2 s	~17× faster
MongoDB	50,000	1.3 s	~19× faster

Verified: `raw` went to **exactly** 2,100,000 / 1,050,000 rows, **0 duplicates** (the delete-then-insert *merge* by PK dedupes correctly) and the watermark advanced. The cost of the sync tracks the **size of the change**, not the size of the base — that is what makes the platform cheap in steady state.

4) Knowledge graph (FalkorDB)

The ontology maps the *clean* layer to a graph: `User`, `Contract` and `Product`, with the edges `PROCESSANTE` and `PROCESSADO` (contract → user) and `SOBRE` (contract → product). Entity resolution is deterministic (same id → same node), so the two parties of each contract point at the correct user nodes.

Why a slice? Materializing the **full** graph (2.1M users + 1.05M contracts ⇒ ~3.15M nodes + ~3.15M edges) would take **~3.3 GB in FalkorDB alone** (extrapolated from the real measurement, ~510 bytes/element) — it does not fit in the laptop's 3.8 GB Docker VM, which also shares RAM with Postgres/Mongo/MinIO. So the graph was materialized over a **representative slice** — 100k contracts + the 190,256 referenced users + 500 products ≈ **291k nodes, 300k edges** — measured, with the rate projected to the full volume. Ingestion and *clean* always ran at **full** volume (2M/1M).

The build was **optimized during the benchmark itself**. It was doing **1 MERGE per row** (one round-trip to FalkorDB per node and per edge); I rewrote the memory-worker's `execute_batch` to **group each batch into a single UNWIND \$rows AS row ...** (1 round-trip per batch of ~500):

Metric	Before (1 MERGE/row)	After (UNWIND)	Full-graph projection
Nodes	290,756	290,756	~3.15M
Edges	300,000	300,000	~3.15M
Build time	362 s	126 s	~22 min
Rate	~1,630 elem/s	~4,680 elem/s	—
Peak RAM (worker)	255 MB	250 MB	—
FalkorDB RAM	302 MB	302 MB	~3.3 GB

2.87× faster, identical result, **0 fallbacks**. The local gain is "only" ~3× because on the same machine the round-trip is cheap — the dominant cost became the `MERGE` itself in FalkorDB + reading the lake. **Against a remote FalkorDB (production, another host), where network latency dominates, UNWIND yields much more.** Next step for the initial load of large graphs: `falkordb-bulk-loader` (CSV → graph, millions/s).

Quality: 100% of edges resolved — PROCESSANTE 100,000/100,000, PROCESSADO 100,000/100,000, SOBRE 100,000/100,000, **0 orphans**. Deterministic entity resolution guarantees the two parties of each contract land on the right user nodes.

A real sub-graph (one contract and its "digital twin") straight from FalkorDB:

```
MATCH (c:Contract)-[:PROCESSANTE]->(a:User),
      (c)-[:PROCESSADO]->(b:User),
      (c)-[:SOBRE]->(p:Product)
RETURN c.contract_id, a.name, b.name, p.product_name, c.status, c.value LIMIT 2
```

contract_id	processante	processado	product	status	value
95152	Usuário 400774	Usuário 5093	Produto 54	closed	42,933.20
88258	Usuário 1968098	Usuário 8993	Produto 485	closed	126,069.85

Digital twin — one contract and its links

Real FalkorDB sub-graph: 1 contract (Mongo) connects its two parties (Postgres) and the product (Mongo)



(Benchmark findings in the graph viz, already fixed: **(a)** first paint requested 5,000 nodes and blew up (HTTP 500) on large graphs — cap lowered; **(b)** first paint fetched only **entity** nodes and, in a hub-and-spoke graph (every link goes through the **contract**), that painted users **floating, with no connection** — "all I see is nodes". Fixed: the first paint now comes from an **edge-seeded sample** (via the read-only Cypher query), so it opens **connected** — the "digital twins" above show up immediately, no clicking required. Per-node expansion still does not return every incoming edge — tracked as a separate follow-up.)

The ontology that generates this graph

The graph above is not hardcoded — it comes from a **declarative map** (`graph_mapping.yaml`):

`User / Product / Contract` nodes (each reading a `clean` table) + `PROCESSANTE / PROCESSADO / SOBRE` edges linking contract → parties and product. Since `postgres / mongodb` are *dynamic-schema* (the schema belongs to the customer), this map is **written per customer** — here, by hand, as the FDE would. Two ways to create/edit it, both writing the **same** thing, versioned:

- **Via the API** (FDE / automation): `PUT /v1/ontology` (saves) + `POST /v1/ontology/apply` (applies) — that is how **Version 4** of this benchmark was registered.
- **Via the UI** (customer): **Settings** → **Ontology** → **YAML tab** → **Edit** → **Save and apply**. The console validates the YAML *and* the columns against the real `clean` tables, saves and applies in one click, with version history for rollback.

strattum.ai

Buscar conectores, memory, knowledge, docs...

acme-prod

Dev User

MENU

Dashboard

CAMADA DE DADOS

Conectores

Transformações

Catálogo de Dados

Lineage

IA / OPS

Memory

Knowledge

Skills

OPERAÇÕES

Health

Pipelines

Evals

Observability

ACESSO A DADOS

Chaves de API

Conexões

Uso

WORKSPACE

Meu Perfil

Times

Usuários

IA & DADOS

Modelos de IA

FinOps

Ontologia

Glossário

Workspace Git

Ontologia

Configure os tipos de entidade, relacionamentos e mapeamento de fontes.

Gráfico

YAML

Versão 4

Atual

1

version: "1"

2

3

Benchmark ontology: Users, Contracts and Products from the DuckLake clean layer.

4

Graph node sources are BOUNDED slices (clean/graph_users, clean/graph_contracts)

5

so the graph fits the 3.8 GB Docker VM; clean.users / clean.contracts hold the

6

full 2M / 1M for the catalog + throughput numbers. Contract parties resolve to

7

Users by the postgres user id (processante_id / processado_id -> user_id).

8

9

nodes:

10

- label: User

11

source: clean/graph_users

12

connector: postgres

13

id_field: user_id

14

er_fields: [email]

15

display_tier: entity

16

display_color: blue

17

properties: [user_id, name, email, cpf, city, updated_at]

18

19

- label: Product

20

source: clean/products

21

connector: mongodb

22

id_field: product_id

23

er_fields: []

24

display_tier: entity

25

display_color: green

26

properties: [product_id, product_name, product_category]

27

28

- label: Contract

29

source: clean/graph_contracts

30

connector: mongodb

31

32

Ver histórico de versões

Footprint, projection and costs at the customer (AWS)

How much the data occupyes (measured)

Where	Size	Note
Lake — raw (Parquet/S3)	171 MB	the 3.15M records, compressed
Lake — clean	174 MB	materialized tables
Graph — FalkorDB (RAM)	302 MB	290k nodes + 300k edges, in-memory (~510 B/elem)
Graph — disk snapshot (RDB)	84 MB	compressed persistence
(comparison) source Postgres	306 MB	the same 2.1M users, OLTP
(comparison) source Mongo	170 MB	the same 1.05M contracts

Two points that become commercial arguments: **(a)** the open Parquet (171 MB) is **more compact than the source operational databases** (306 + 170 MB) — the data stays yours, queryable by any engine, with no proprietary copy; **(b)** ingestion and clean have a **constant** RAM peak (272–394 MB) *regardless of volume* — streaming, they never load the dataset.

What this handles on a real server

The benchmark ran on an 8 GB laptop on purpose. On a server, each stage scales differently — and the bottleneck is **not** where most people think.

STRATTUM — CONFIDENCIAL

pág. 6 / 14

Lake (Parquet on S3) — practically unlimited, and it costs cents. At ~56 bytes/row compressed (measured: 3M rows → 171 MB), the *file* storage is the cheapest part of the bill. S3 Standard price per GB·month: **\$0.023** in us-east-1 · **\$0.0405** in São Paulo (sa-east-1):

Rows	Parquet in the lake	S3 us-east-1	S3 São Paulo
100 million	~5.6 GB	~\$0.13/mo	~\$0.23/mo
1 billion	~56 GB	~\$1.30/mo	~\$2.27/mo
10 billion	~560 GB	~\$13/mo	~\$23/mo

The DuckLake catalog adds no cost of its own: it is ~30 metadata tables in the **same Postgres** the platform already runs. S3 requests (PUT/GET) are negligible under this access pattern — ingestion writes micro-batch files (a few thousand PUTs per sync, at \$0.005/1k) and reads are columnar scans, not millions of small GETs.

Ingestion — linear in time, constant RAM (~48.5k rows/s, ~272 MB fixed → never OOMs): 100M rows ≈ **~34 min**, 1B rows ≈ **~5.7 h**. With more cores / parallel connectors / faster S3, it shortens.

Clean (dbt/DuckDB) parallelizes per core and spills to disk — 100M rows in ~a few minutes on an ordinary server.

Incremental — the secret of the low cost. The day-to-day sync costs the **size of the change**, not of the base. A customer with 1 billion rows but 5 million changing per day pays for ~5M/day (~2 min) — **not** 1 billion. In steady state, the bill always tracks the delta.

Graph (FalkorDB) — RAM-bound, but a normal server holds tens to hundreds of millions of entities (at the measured ~510 B/element, with ~40% headroom for query RAM):

Server RAM	Elements in the graph	≈ Entities (nodes)
16 GB	~19M	~9M
32 GB	~38M	~19M
64 GB	~75M	~37M
128 GB	~150M	~75M
256 GB	~300M	~150M

FalkorDB officially scales to **billions of edges** on adequate hardware, with millisecond queries (GraphBLAS). The practical limit of the **initial load** is write speed — today ~4,680 elem/s (with UNWIND); for large graphs, `falkordb-bulk-loader` does millions/s. After that, the worker maintains the graph **by delta** (cheap).

What it costs to keep the graph in RAM (AWS)

RAM is the only resource the graph truly consumes — and on AWS you buy it with the instance. For graphs that still fit in the platform VM, the extra cost is **zero**: on the 16 GB starter VM, FalkorDB gets a ~4 GB slice (the compose cgroup ceiling) → **up to ~4–5M elements at no additional cost**. Beyond that, the graph moves to a **dedicated memory-optimized node** (`r8g` family, Graviton4 — 8 GB of RAM per vCPU, the best \$/GB on AWS). On-demand Linux prices (Jul/2026):

Elements in the graph	RAM (with query headroom)	Instance	us-east-1	São Paulo
up to ~5M	slice of the starter VM	— (shared)	\$0 extra	\$0 extra
~19M	16 GB	r8g.large	~\$86/mo	~\$162/mo
~38M	32 GB	r8g.xlarge	~\$172/mo	~\$324/mo
~75M	64 GB	r8g.2xlarge	~\$344/mo	~\$647/mo
~150M	128 GB	r8g.4xlarge	~\$688/mo	~\$1,294/mo

In other words: **graph RAM costs ~\$5.4/GB-month in us-east-1 and ~\$10/GB-month in São Paulo** (on-demand; a 1-year savings plan cuts ~35%). Graph persistence (RDB snapshot + AOF) goes to the EBS disk and is irrelevant in the bill — this benchmark's 590k elements produced an **84 MB** snapshot (~\$0.01/mo on gp3). The operational rule we already apply in compose holds here too: the instance's RAM ceiling must stay **above** FalkorDB's `maxmemory`, so the BGSAVE/AOF-rewrite fork has headroom.

The platform VM (the rest of the bill)

The rest of the stack (ingestion, dbt, APIs, Prefect, Postgres, Qdrant) runs on the starter VM — `t4g.xlarge` (4 vCPU / 16 GB, Graviton2, ADR-005): **~\$98/mo** in us-east-1 · **~\$157/mo** in São Paulo, plus the EBS gp3 disk ($\$0.08 \cdot \0.152 per GB-month — 100 GB $\approx \$8 \cdot \15). Ingestion and clean are **streaming with constant RAM** (\$1–2), so the same VM serves from 3M to hundreds of millions of rows — the time grows, the machine does not.

Example — a typical customer (2M customers + 8M contracts + ~100k changes/day), full AWS bill in São Paulo:

- **Lake (files):** ~600 MB of Parquet → **~\$0.03/mo** of S3.
- **Initial ingestion** ~3–4 min · **clean** ~1 min (on the VM itself).
- **Graph:** ~10M nodes + ~24M edges $\approx$ 34M elements → 48 GB with headroom → dedicated `r8g.2xlarge` node (64 GB) → **~\$647/mo** (us-east-1: ~\$344); initial load via bulk-loader in minutes.
- **Platform VM:** `t4g.xlarge` + 100 GB gp3 → **~\$172/mo** (us-east-1: ~\$106).
- **Daily sync:** ~100k rows → **seconds** (the delta is the permanent bill, and it is tiny).

Total AWS infra: ~\$820/mo in São Paulo (~\$450 in us-east-1), on-demand — and the dominant item is graph RAM, which only grows if the *graph* grows. No proprietary warehouse, no cluster, all BYOC in the customer's own account.

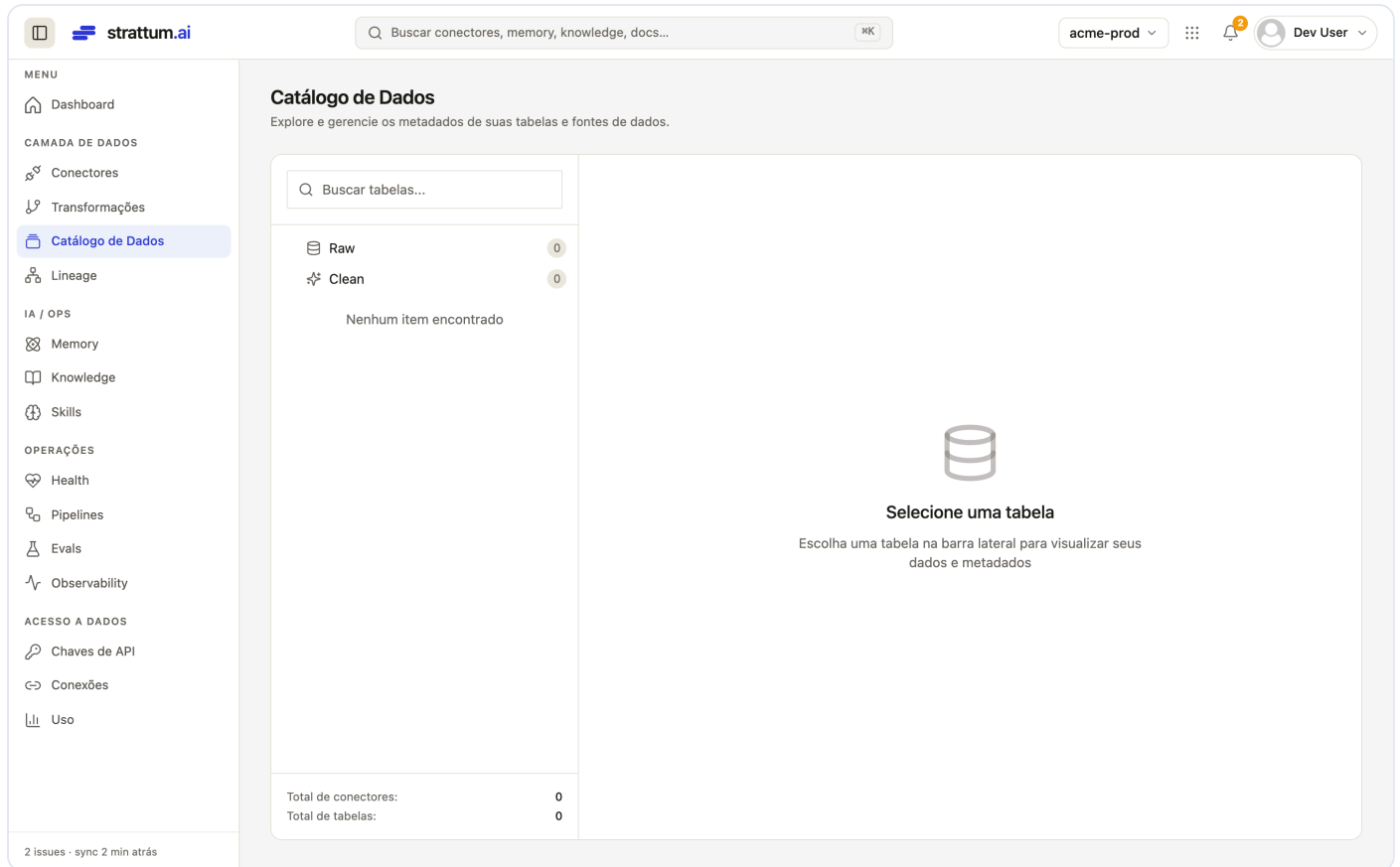
Everything on a single **BYOC** server (in the customer's own cloud), with no proprietary data warehouse. **What scales to billions (lake, ingestion, incremental) is cheap; what is RAM-bound (the graph) grows with the server — and the permanent cost is always the delta's.**

Price sources (Jul/2026, on-demand Linux): official AWS price table — `r8g.large` sa-east-1 \$0.2216/h and `t4g.large` sa-east-1 \$0.1072/h ($\Rightarrow$ `t4g.xlarge` \$0.2144/h); us-east-1: `t4g.xlarge` \$0.1344/h · `r8g.large` \$0.1178/h; EBS gp3 sa-east-1 \$0.152/GB-month. Monthly = hourly $\times$ 730. S3 Standard sa-east-1 \$0.0405/GB-month (first 50 TB) — double-check in the AWS calculator when closing a proposal; no BRL conversion applied.

Before → After (Strattum console)

The whole journey is visible in the console — no hand-written SQL.

Before — empty catalog, empty memory:



strattum.ai

Buscar conectores, memory, knowledge, docs...

acme-prod

Dev User

MENU

Dashboard

CAMADA DE DADOS

Conectores

Transformações

Catálogo de Dados

Lineage

IA / OPS

Memory

Knowledge

Skills

OPERAÇÕES

Health

Pipelines

Evals

Observability

ACESSO A DADOS

Chaves de API

Conexões

Uso

2 issues - sync 2 min atrás

Memory

Memória episódica de longo prazo da organização.

NER Review

Nenhuma entidade corresponde aos filtros atuais.

After — 3M rows cataloged + graph materialized:

Catalog: 3 raw tables + 5 clean = 8 total (was 0), with nobody writing ETL SQL.

strattum.ai

Buscar conectores, memory, knowledge, docs...

acme-prod

Dev User

MENU

Dashboard

CAMADA DE DADOS

Conectores

Transformações

Catálogo de Dados

Lineage

IA / OPS

Memory

Knowledge

Skills

OPERAÇÕES

Health

Pipelines

Evals

Observability

ACESSO A DADOS

Chaves de API

Conexões

Uso

2 issues - sync 2 min atrás

Catálogo de Dados

Explore e gerencie os metadados de suas tabelas e fontes de dados.

Buscar tabelas...

Raw

3

MongoDB

2

PostgreSQL

1

Clean

5

Selezione una tabela

Escolha uma tabela na barra lateral para visualizar seus dados e metadados

Total de conectores: 2

Total de tabelas: 8

STRATTUM — CONFIDENCIAL

pág. 10 / 14

Clean layer with the real counts — users 2,100,000, contracts 1,050,000, products 500, plus the graph slices:

strattum.ai

Buscar conectores, memory, knowledge, docs...

acme-prod

Dev User

MENU

Dashboard

CAMADA DE DADOS

Conectores

Transformações

Catálogo de Dados

Lineage

IA / OPS

Memory

Knowledge

Skills

OPERAÇÕES

Health

Pipelines

Evals

Observability

ACESSO A DADOS

Chaves de API

Conexões

Uso

Catálogo de Dados

Explore e gerencie os metadados de suas tabelas e fontes de dados.

Buscar tabelas...

Raw 3

MongoDB 2

PostgreSQL 1

Clean 5

contracts 1050k

graph_contracts 100k

graph_users 190k

products 500

users 2100k

Total de conectores: 2

Total de tabelas: 8

users

8 campos 2.100.000 linhas

Dados Estrutura

Buscar nas linhas...

Arquivo: lake://clean.users — exibindo 50 de 2.100.000 linhas

USER_ID	EXTERNAL_ID	NAME	EMAIL	CPF	CITY
299	299	Usuário 299	user299@example.com	00000000299	Curitiba
599	599	Usuário 599	user599@example.com	00000000599	Fortalez
899	899	Usuário 899	user899@example.com	00000000899	Curitiba
1199	1199	Usuário 1199	user1199@example.com	00000001199	Fortalez
1499	1499	Usuário 1499	user1499@example.com	00000001499	Curitiba
1799	1799	Usuário 1799	user1799@example.com	00000001799	Fortalez
2099	2099	Usuário 2099	user2099@example.com	00000002099	Curitiba
2399	2399	Usuário 2399	user2399@example.com	00000002399	Fortalez
2699	2699	Usuário 2699	user2699@example.com	00000002699	Curitiba
2999	2999	Usuário 2999	user2999@example.com	00000002999	Fortalez
3299	3299	Usuário 3299	user3299@example.com	00000003299	Curitiba

Página 1 de 42000 · linhas 1-50 de 2.100.000

< Anterior

Próximo >

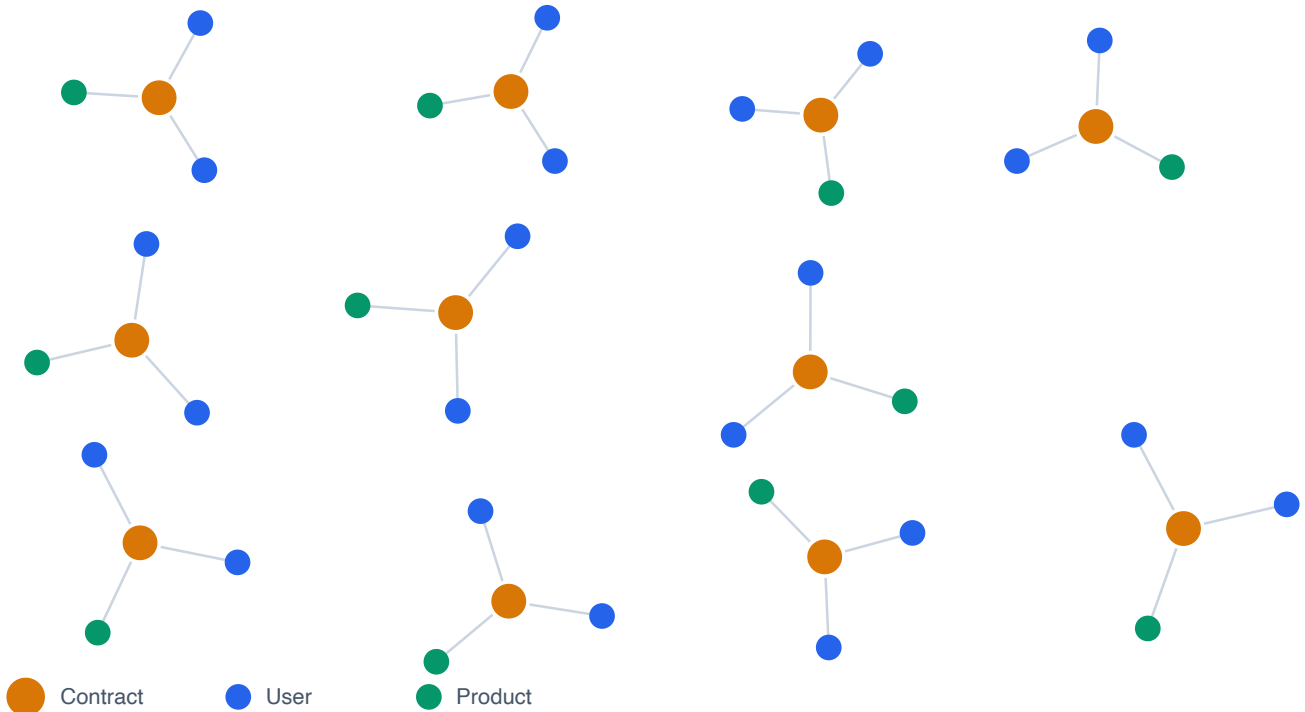
Entity graph in Memory — the first paint now opens **connected**: an edge-seeded sample (~800 nodes) brings the **contracts** (amber) linking their two parties — the **users** (blue) — and the **product** (green). They are §4's "digital twins", repeated ~200x. The pattern's schema:

STRATTUM — CONFIDENCIAL

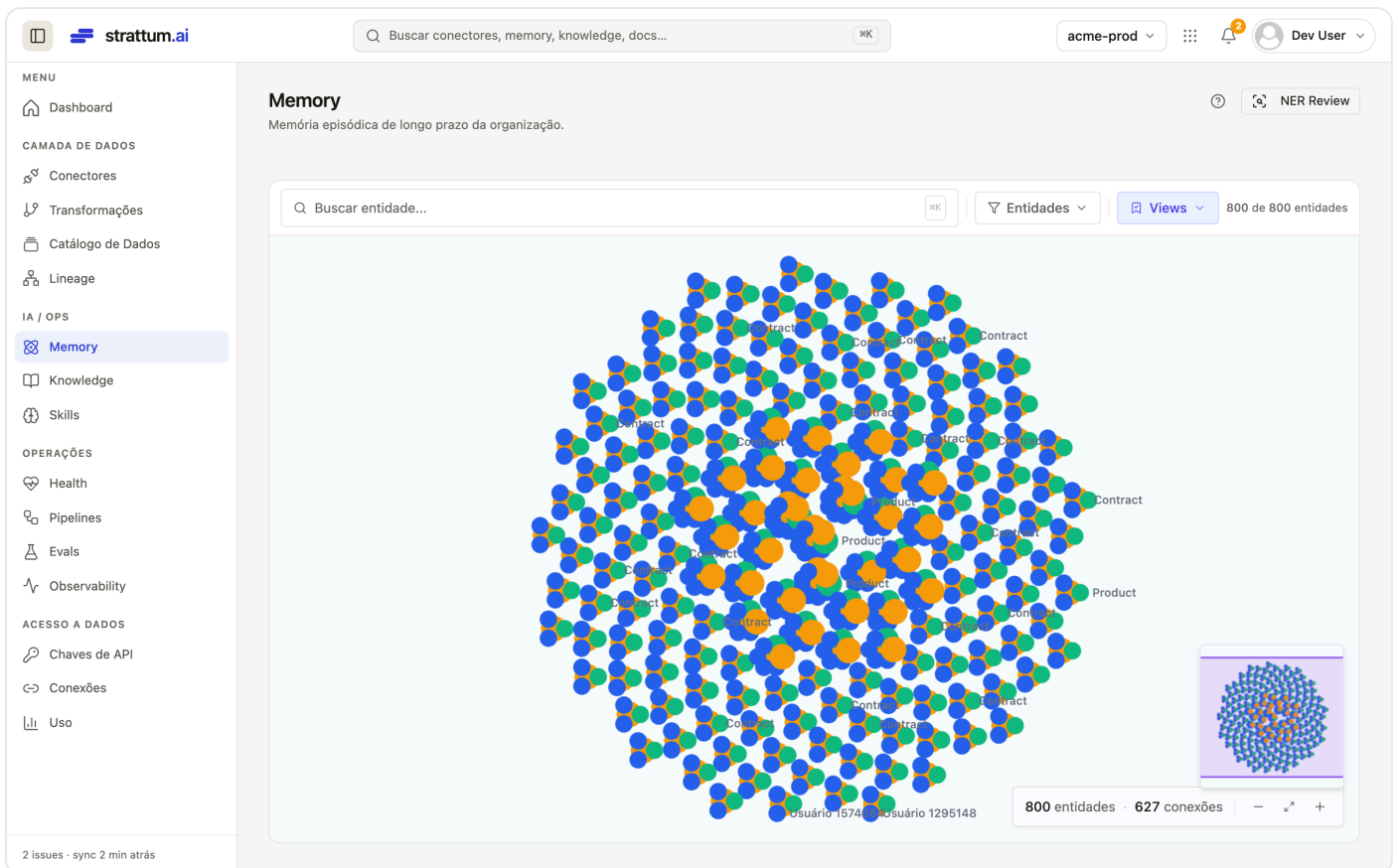
pág. 11 / 14

First paint — now connected

Edge-seeded sample: each contract (●) links its two parties (users) and the product — the "digital twins", repeated.



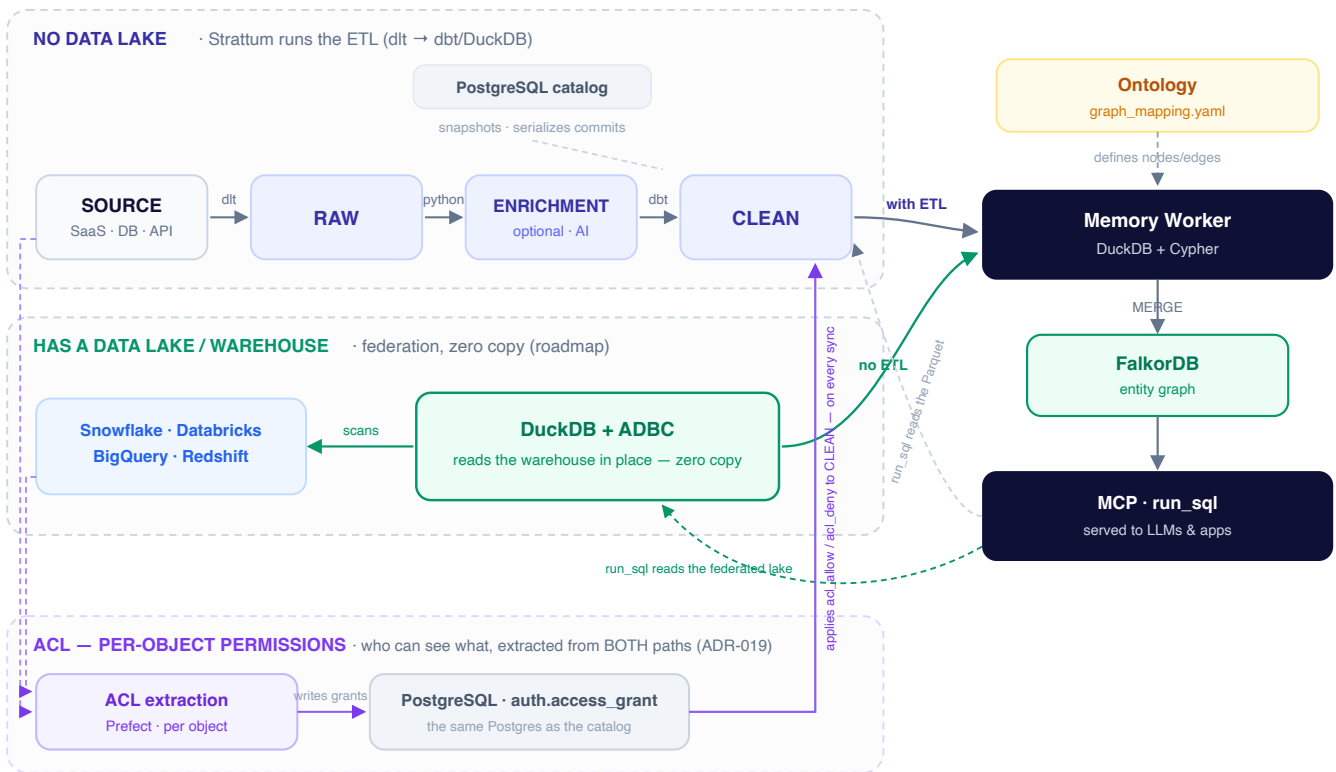
And the same **live in the console** — note the footer: `800 entities • 627 connections` (before: 0 connections):



How it works (4 steps)

How data comes in, becomes a graph, and is served

Two ingestion paths → one knowledge graph → served over SQL and MCP



1. **Connect** the source in the console (PostgreSQL, MongoDB, ...). Credentials live in the connector registry; table/collection selection and sync mode (full/incremental, cursor, PK) live in the connector state.
2. **Ingest** into the lake's `raw` — Parquet on S3/MinIO, cataloged by **DuckLake**. Streaming end to end: constant memory, cursor-based incremental.
3. **Model** with **dbt** running native DuckDB over the lake: normalization, `JOIN`s, typing → `clean` layer, materialized back into the lake itself.
4. **Build the graph**: the ontology maps `clean` → nodes and edges; the worker resolves entities and materializes into FalkorDB. The result shows up in the console's Memory.

Everything on **open formats** (Parquet + a relational catalog). No warehouse lock-in, no proprietary copy of the data.

Why this matters

- **Any company can do it.** If it runs on an 8 GB laptop with 3M rows in ~1 minute, it runs in your environment with your data.
- **Cheap in steady state.** The daily sync costs the **size of the delta**, not of the base.
- **Open by construction.** Parquet + DuckLake: the data stays yours, queryable by any engine, with no re-import.
- **From raw data to knowledge.** It does not stop at the "data lake": it delivers a modeled layer + an entity graph ready for AI/context.

Methodology & reproducibility

- **Measurement:** each stage ran as an isolated process under `/usr/bin/time -l` (wall time + *maximum resident set size*). Throughput = rows ÷ stage time.
- **Data:** synthetic and deterministic (versioned generators). The increment's `updated_at` is strictly greater than the bootstrap watermark, to exercise a real delta.
- **Production code:** ingestion uses the platform's real connectors/flows (`sync_resource_to_raw`), not a separate benchmark path. The `clean` uses the same dbt models. The graph uses the same `MemoryWorkerPipeline`.
- **Scope honesty:** ingestion and clean at full volume (2M/1M); graph on a representative slice with projection, due to the laptop's RAM limit (see §4).
- **The benchmark doubled as QA:** as a bonus, it exposed (and this report fixed) the graph write bottleneck — 1 `MERGE` /row → batched `UNWIND` , **2.87× faster** — and the viz first-paint cap that blew up on large graphs.

Benchmark executed on 2026-07-19; graph optimized (UNWIND) the same day.